

TEREK JOHNSON

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EDUCATION

Master of Science, Computer Science, University of California, Riverside Expected 2027

Bachelor of Science, Computer Science, University of California, Riverside GPA 3.51, June 2026

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Embedded Systems, Operating Systems, Computer Architecture

WORK EXPERIENCE

Software Engineering Intern June 2026 – Aug 2026
NASA Jet Propulsion Laboratory (JPL) *Pasadena, CA*

- Author Kaitai Struct (KSY) specifications from Deep Space Network configuration files to parse/serialize nested binary telemetry, unsupported by JPL's legacy Telemetry Tracking & Command (TTC) software.
- Build a lightweight parsing library, easier to install/maintain than TTC, for JPL's project data system engineers.

Transfer Student Peer Mentor Sep 2025 – June 2026
Marlan & Rosemary Bourns College of Engineering, University of California, Riverside *Riverside, CA*

- Directed social media/marketing campaigns, growing non-follower engagement 8%; advised transfer/prospective students on academics and careers; supervised study spaces and represented the college at outreach events.

TECHNICAL PROJECTS

Single-Axis Solar Tracker & Wireless Transceiver Network Feb 2026

- Engineered a single-axis solar tracker using dual lux sensors, an Arduino Nano, and an H-bridge-driven DC motor to orient a photovoltaic panel toward peak sunlight for a solar-powered boat competition.
- Built a wireless transceiver system integrating a GPS module, an IMU, and power and lux sensors to transmit real-time telemetry from the lake to a shore-side station, enabling live monitoring during race conditions.
- Coordinated design and integration across mechanical, electrical, and software subteams.

Machine Learning Model Analysis on Titanic Dataset Feb 2026

- Applied one-nearest-neighbor forward/backward feature-elimination to identify the optimal feature subset for a passenger survival-prediction model, then diagnosed a critical training error, improving prediction accuracy by 30%.

Full-Life Mar 2025 – June 2025

- Designed and built the frontend of a lifestyle-planning web application using Figma, React, HTML, and CSS, integrating OpenWeather and SERP APIs to generate personalized activity recommendations.
- Containerized the application with Docker to simplify multi-service setup and deployment.

RESEARCH EXPERIENCE

Undergraduate Research Assistant June 2024 – Sep 2024
Collaborative Intelligent Systems Lab, University of California, Riverside *Riverside, CA*

- Selected as a 2024 Bourns College of Engineering Summer Bridge Scholar; collected and synchronized a multi-modal LiDAR/camera dataset with a team of eight, using instrumented vehicles and city sensors to support autonomous-driving ML pipelines.
- Evaluated data quality and debugged sensor drivers, validating the dataset for training/benchmarking ML models; presented findings at the 2024 Bourns College of Engineering Summer Research Symposium.

SKILLS & HONORS

Languages: C++, C, C#, Java, JavaScript, Python

Frameworks & Tools: React, HTML, CSS, TailwindCSS, SFML, Docker, MongoDB, Git/GitHub, Agile/Scrum, CI/CD

Honors: University Regents Scholar (2024) • Dean's Honor List • Southern California Edison Scholarship (2025)